# Intro to Computational Linguistics

# Contest 1: Aspect-based Sentiment Analysis

Due date 7 March 2026 (no late submission allowed)

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## Task Overview

Your goal is to build an aspect-based sentiment analysis (ABSA) system that can identify both aspects and sentiments in text. This system needs two main components: (1) an aspect classifier that finds the topics being discussed, and (2) a sentiment classifier that determines whether the opinion about each topic is positive, negative, or neutral. For your prediction to be marked correct, both the aspect and sentiment must match the expected answer.

## How You'll Be Graded

You can work on this assignment alone or with one partner. Important: No late submissions will be accepted since we need to compare everyone's results fairly across the class.

Your assignment will be evaluated based on two equally important criteria:

- System Performance (50%): How well your model performs, measured by Overall F1-score compared against the other submissions.
- Report (50%)
    - Process and Analysis (30%): The quality of your approach and insights, documented in your written report. The report should demonstrate an understanding of the principles for improving model accuracy and the factors affecting model accuracy.
    - Quality of writing (20%): must be easy to read and clear. The tables and figures must be well-formatted.

You can use any approach for your final system, but you must test and compare at least two of these baseline models:

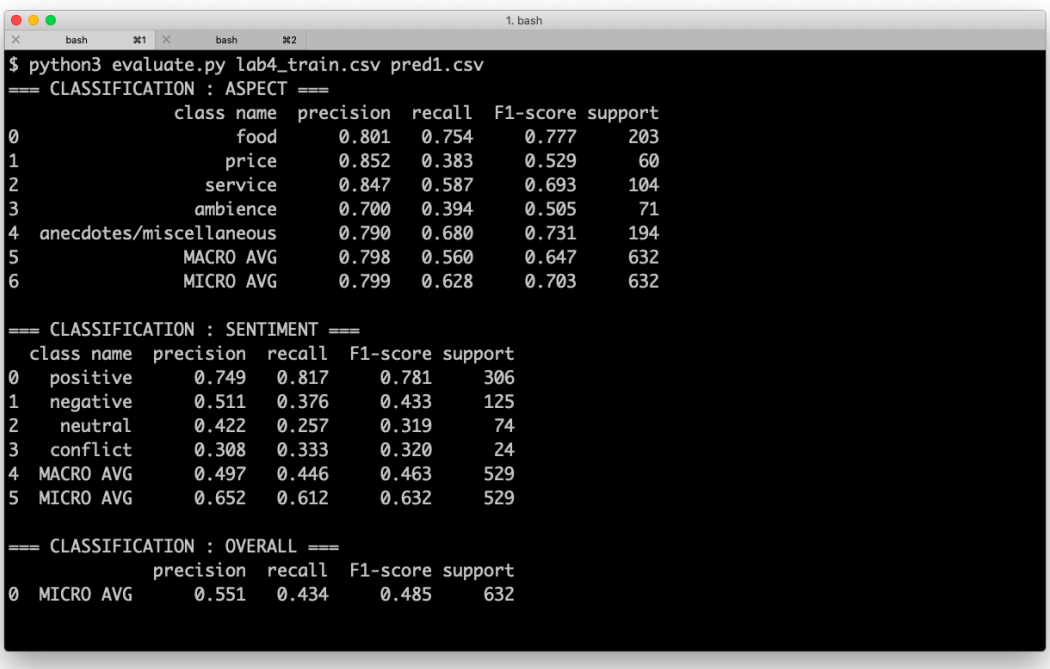
1. Logistic Regression: Using bag-of-words features (a classic machine learning approach)
2. Deep Averaging Network: A neural network that averages word embeddings
3. Zero-shot LLM Classifier: Using a pre-trained language model without additional training
4. Fine-tuned BERT Classifier: Fine-tune any pretrained encoder-only language model with training data

Your report should document all the models you tried, including the ones that didn't work as well. This helps us understand your thought process and what you learned. Make sure to include enough detail so that someone could understand and potentially reproduce your work.

## Provided materials:

1. Dataset which can be divided into training set and dev set as desired.
2. Test set which will not yet have labels.
3. evaluate.py for calculating Overall F1 score – a prediction will be considered correct only if both aspect and sentiment are correct. Example usage (must use terminal or anaconda prompt):

```
>>> python evaluate.py contest1_train.csv (filename).csv
```



```
$ python3 evaluate.py lab4_train.csv pred1.csv
=== CLASSIFICATION : ASPECT ===
                class name  precision  recall  F1-score support
0                     food      0.801   0.754     0.777     203
1                    price      0.852   0.383     0.529      60
2                  service      0.847   0.587     0.693     104
3                 ambience      0.700   0.394     0.505      71
4  anecdotes/miscellaneous      0.790   0.680     0.731     194
5                MACRO AVG      0.798   0.560     0.647     632
6                MICRO AVG      0.799   0.628     0.703     632

=== CLASSIFICATION : SENTIMENT ===
  class name  precision  recall  F1-score support
0   positive      0.749   0.817     0.781     306
1   negative      0.511   0.376     0.433     125
2    neutral      0.422   0.257     0.319      74
3   conflict      0.308   0.333     0.320      24
4  MACRO AVG      0.497   0.446     0.463     529
5  MICRO AVG      0.652   0.612     0.632     529

=== CLASSIFICATION : OVERALL ===
              precision  recall  F1-score support
0  MICRO AVG      0.551   0.434     0.485     632
```

4. check_id.py for checking the IDs of the test_pred.py we predicted, to see if all IDs have been predicted/if there are any strange IDs, by comparing them with the IDs in contest1_test.py.

```
>>> python check_id.py contest1_test.csv test_pred.csv
```

# What to submit:

1. **test_pred.csv** :Submit a prediction csv of the test set created from a DataFrame with columns similar to the training set: 'id', 'aspectCategory', 'polarity'. Then save it into a file named test_pred.csv. Before submitting, check if the model has been run for all IDs. Note: If a text has more than 1 aspect, add an additional row but keep the text's ID. Therefore, the submitted csv may have more rows than the original test set csv, for example:

| id | aspectCategory | polarity |
|---|---|---|
| 123 | food | positive |
| 368 | service | negative |
| 368 | price | positive |

(The test set csv only has id and text, and the id values are not unique like the train set because there is no gold standard label)

2. **report.pdf :**Your written report should be no longer than five pages and should present a clear account of your experiments and your analysis of the results. Think of it as a research-style paper: the goal is not only to show what you did, but also to demonstrate that you understood the task, designed the experiments carefully, and interpreted the outcomes critically.

   **Note on the use of AI**: You are welcome to use an AI assistant of your choice to help you write up the report. But you must make sure that the model details, figures, and numbers are accurate and be sure to form your own idea of what you observe in the experiment and what you personally learn from this assignment in the conclusion.

## *Experimental Setup*

Begin by explaining how you formulated the classification task. Was it framed as a multi-class problem, a binary decision, or a more complex formulation such as joint aspect–polarity classification? Motivate your choice and explain what makes this formulation appropriate for the dataset and research question.

Next, describe how you split your data into training and validation sets. Report the label distribution for the split in a table so that the reader can see whether the dataset is balanced or skewed.

## *Model*

This section should describe the models you experimented with. Include both the models that worked well and those that did not perform as expected. For each model, provide enough details so that someone could in principle reproduce your setup. This includes the type of features used, any preprocessing steps, choices about embeddings (whether they were pre-trained, fine-tuned, or learned from scratch), and the key hyperparameters such as number of layers, hidden units, learning rate, and batch size. Justify these decisions: why did you expect a certain model to work better than another? What trade-offs did you consider? Be clear about your reasoning so that the report goes beyond “we tried X and Y” to “we tried X because we expected it to capture ___, but Y might perform better because ___.”

## *Results*

Present your experimental results in a clear comparison table. Every model you tested should appear here so that the reader has a complete picture of the progression of your experiments. Report evaluation metrics that are appropriate for the task (e.g., accuracy, precision, recall, F1 for every class, and macro/micro averages) and explain why those metrics are informative. Alongside the table, provide a narrative analysis: which models performed best overall, which struggled, and what factors likely contributed to these outcomes? Try to connect the performance differences back to your earlier discussion of model design, hyperparameters, and data distribution.

## *Conclusion*

Conclude your report by summarizing what you have learned from your experiments. Indicate clearly which model you selected for the test set submission to the contest, and explain why. Your reasoning should link back to the results section, highlighting both quantitative evidence (performance metrics) and qualitative insights (interpretability, robustness, or computational efficiency).